DA5000W Linear algebra and optimization

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Tutorial 4 Linear algebra

1 Least Squares

Least square method is the process of finding a regression line or best-fitted line for any data set that is described by an equation.

It addresses the question,

> If $Ax = b$ does not have a solution, what is the best approximate solution?

The best approximate solution is called the least squares solution to the inconsistent system of equations.

1.1 Least squares solution

Let A be an $m \times n$ matrix and let $\mathbf{b} \in \mathbb{R}^m$. A least-squares solution of the matrix equation $A\mathbf{x} = \mathbf{b}$ is a vector $\hat{\mathbf{x}} \in \mathbb{R}^n$ such that

$$\|A\hat{\mathbf{x}} - \mathbf{b}\| \leq \|A\mathbf{x} - \mathbf{b}\| \quad \text{for all } \mathbf{x} \in \mathbb{R}^n.$$

Here, $\|A\mathbf{x} - \mathbf{b}\|$ represents the *distance* between the vectors $A\mathbf{x}$ and $\mathbf{b}$

Hence, a least-squares solution minimizes the *sum of the squares of the differences* between the components of $\mathbf{b}$ and those of $A\mathbf{x}$.

1.2 Theorem: 1.1

Let A be an $m \times n$ matrix and let $\mathbf{b}$ be a vector in $\mathbb{R}^m$. The least-squares solutions of $A\mathbf{x} = \mathbf{b}$ are the solutions of the matrix equation

$$A^T A\mathbf{x} = A^T\mathbf{b}.$$

1.3 Computing a Least squares solution

1. Compute the matrix $A^T A$ and the vector $A^T\mathbf{b}$.
2. Form the augmented matrix for the system $A^T A\mathbf{x} = A^T\mathbf{b}$, and row-reduce.
3. This system is always consistent, and any solution $\hat{\mathbf{x}}$ is a least-squares solution.

1.4 Theorem 1.2

The following statements are equivalent:

1. The equation $A\mathbf{x} = \mathbf{b}$ has a unique least-squares solution.
2. The columns of A are linearly independent.
3. The matrix $A^T A$ is invertible.

In this case, the least-squares solution is given by

$$\hat{\mathbf{x}} = (A^T A)^{-1} A^T\mathbf{b}.$$

1.5 Error Analysis: The Sum of Squared Errors (SSE)

The goal is to minimize the total squared difference between the observed data points and the predicted values from the model.

Let the observed data be

$$(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n),$$

and let the predicted values from the fitted model be $\hat{y}_i$.
Then the **sum of squared errors (SSE)** is defined as:

$$\boxed{E = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2}$$

1.6 Examples

Consider the four data points,

$$(1, 2), \quad (2, 2.5), \quad (3, 4), \quad (4, 3.5).$$

1.6.1 Linear fit: $y = mx + c$

Matrix form: $A\mathbf{x} = \mathbf{b}$ with

$$A = \begin{pmatrix} 1 & 1 \\ 2 & 1 \\ 3 & 1 \\ 4 & 1 \end{pmatrix}, \qquad \mathbf{x} = \begin{pmatrix} m \\ c \end{pmatrix}, \qquad \mathbf{b} = \begin{pmatrix} 2 \\ 2.5 \\ 4 \\ 3.5 \end{pmatrix}.$$

Normal equations:

$$A^T A = \begin{pmatrix} 30 & 10 \\ 10 & 4 \end{pmatrix}, \qquad A^T \mathbf{b} = \begin{pmatrix} 33 \\ 12 \end{pmatrix},$$

so

$$\begin{pmatrix} 30 & 10 \\ 10 & 4 \end{pmatrix} \begin{pmatrix} m \\ c \end{pmatrix} = \begin{pmatrix} 33 \\ 12 \end{pmatrix}.$$

Solving gives (exact)

$$m = \frac{3}{5} = 0.6, \qquad c = 3 - 2.5(0.6) = 1.5$$

Thus the least-squares line is

$$\boxed{\hat{y}_{\text{lin}} = 0.6x + 1.5}.$$

Predicted values, residuals and SSE:

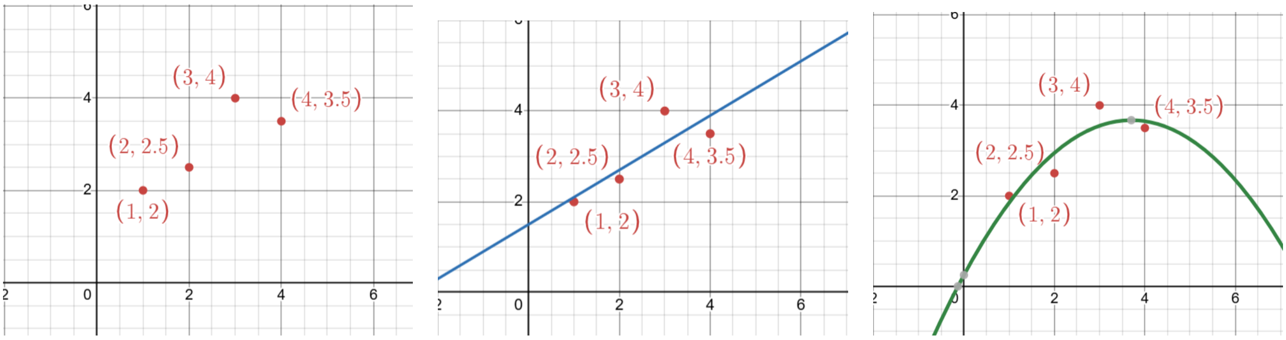
x	y_{obs}	$y_{\text{pred}} = \frac{4}{5}x + \frac{13}{10}$	$e_i = y_{\text{obs}} - y_{\text{pred}}$
1	2.0	2.1	-0.1
2	2.5	2.7	-0.2
3	4.0	3.3	$+0.7$
4	3.5	3.9	-0.4

$$\text{SSE}_{\text{lin}} = (-0.1)^2 + (-0.2)^2 + (0.7)^2 + (-0.4)^2 = 0.7.$$

1.6.2 Quadratic fit: $y = ax^2 + bx + c$

Matrix form: $A\mathbf{x} = \mathbf{b}$ with

$$A = \begin{pmatrix} 1^2 & 1 & 1 \\ 2^2 & 2 & 1 \\ 3^2 & 3 & 1 \\ 4^2 & 4 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 \\ 4 & 2 & 1 \\ 9 & 3 & 1 \\ 16 & 4 & 1 \end{pmatrix}, \qquad \mathbf{x} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}, \qquad \mathbf{b} = \begin{pmatrix} 2 \\ 2.5 \\ 4 \\ 3.5 \end{pmatrix}.$$



(a) Data points

(b) Linear fit

(c) Quadratic fit

Figure 1: Comparison of least-squares fits: data points, linear model, and quadratic model.

Compute the normal system $A^T A\mathbf{x} = A^T\mathbf{b}$:

$$A^T A = \begin{pmatrix} 354 & 100 & 30 \\ 100 & 30 & 10 \\ 30 & 10 & 4 \end{pmatrix}, \qquad A^T\mathbf{b} = \begin{pmatrix} 104.0 \\ 33.0 \\ 12.0 \end{pmatrix}.$$

Solving yields the exact coefficients

$$a = -\frac{1}{4}, \qquad b = \frac{37}{20} = 1.85, \qquad c = \frac{1}{4}.$$

Thus the least-squares quadratic is

$$\boxed{\hat{y}_{\text{quad}} = -\tfrac{1}{4}x^2 + \tfrac{37}{20}x + \tfrac{1}{4}}.$$

Predicted values, residuals and SSE:

x	y_{obs}	$y_{\text{pred}} = -0.25x^2 + 1.85x + 0.25$	$e_i = y_{\text{obs}} - y_{\text{pred}}$
1	2.0	1.85	+0.15
2	2.5	2.95	−0.45
3	4.0	3.55	+0.45
4	3.5	3.65	−0.15

$$\text{SSE}_{\text{quad}} = (0.15)^2 + (-0.45)^2 + (0.45)^2 + (-0.15)^2 = 0.45.$$

The same four points produce $\text{SSE}_{\text{lin}} = 0.7$ for the best line and $\text{SSE}_{\text{quad}} = 0.45$ for the best quadratic; the quadratic fits these points better (smaller SSE).

2 Orthogonality and Orthonormal Bases

2.1 Definition: Orthogonality

Two vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ are said to be **orthogonal** if

$$\mathbf{u} \cdot \mathbf{v} = 0.$$

A set of vectors $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n\}$ is called an **orthogonal set** if every pair of distinct vectors is orthogonal:

$$\mathbf{u}_i \cdot \mathbf{u}_j = 0 \quad \text{for all } i \neq j.$$

If, in addition, each vector has unit length ($\|\mathbf{u}_i\| = 1$), then the set is called an **orthonormal basis**.

2.2 Example

Consider the orthonormal basis for $\mathbb{R}^3$ and the orthonormal set:

$$\mathbf{u}_1 = \frac{1}{\sqrt{2}}(1, 1, 0), \quad \mathbf{u}_2 = \frac{1}{\sqrt{3}}(1, -1, 1), \quad \mathbf{u}_3 = \frac{1}{\sqrt{6}}(-1, 1, 2).$$

Let us find the coordinates of a vector$\mathbf{v} = (2, 1, 3)$ wrt this orthonormal basis.

2.2.1 Step 1: Compute the coefficients

$$c_i = \mathbf{u}_i \cdot \mathbf{v}.$$

$$c_1 = \frac{1}{\sqrt{2}}(2+1+0) = \frac{3}{\sqrt{2}}, \quad c_2 = \frac{1}{\sqrt{3}}(2-1+3) = \frac{4}{\sqrt{3}}, \quad c_3 = \frac{1}{\sqrt{6}}(-2+1+6) = \frac{5}{\sqrt{6}}.$$

2.2.2 Step 2: Express v as a linear combination

$$\mathbf{v} = c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + c_3\mathbf{u}_3 = \frac{3}{2}(1,1,0) + \frac{4}{3}(1,-1,1) + \frac{5}{6}(-1,1,2).$$

2.3 Remark

If the basis were not orthonormal, finding these coefficients would require solving a system of equations:

$$A\mathbf{c} = \mathbf{v}, \quad \text{where } A = [\mathbf{u}_1\ \mathbf{u}_2\ \mathbf{u}_3].$$

But since the basis is orthonormal,$A^TA = I$, and therefore the coefficients are obtained simply by

$$\boxed{\mathbf{c} = A^T\mathbf{v}.}$$

This property dramatically simplifies computations, especially in projections and least-squares problems.

2.4 Geometric Interpretation

Each coefficient c_i represents the projection of $\mathbf{v}$ onto the corresponding basis vector $\mathbf{u}_i$.

Hence, the components are simply the lengths of the projections.

3 The Gram–Schmidt Orthonormalization Process

3.1 Motivation

Theoretical Description

Let $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ be a linearly independent set in $\mathbb{R}^m$.

We construct an orthogonal set $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n\}$ as follows:

$$\mathbf{u}_1 = \mathbf{v}_1,$$

$$\mathbf{u}_2 = \mathbf{v}_2 - \text{proj}_{\mathbf{u}_1}(\mathbf{v}_2) = \mathbf{v}_2 - \frac{\mathbf{v}_2 \cdot \mathbf{u}_1}{\mathbf{u}_1 \cdot \mathbf{u}_1}\,\mathbf{u}_1,$$

$$\mathbf{u}_3 = \mathbf{v}_3 - \text{proj}_{\mathbf{u}_1}(\mathbf{v}_3) - \text{proj}_{\mathbf{u}_2}(\mathbf{v}_3), \quad \text{and so on.}$$

After obtaining the orthogonal set, we normalize each vector to obtain an orthonormal set:

$$\mathbf{e}_i = \frac{\mathbf{u}_i}{\|\mathbf{u}_i\|}, \quad i = 1, 2, \ldots, n.$$

3.2 Example

$$\mathbf{v}_1 = (1,1,0), \quad \mathbf{v}_2 = (1,0,1), \quad \mathbf{v}_3 = (0,1,1).$$

3.2.1 Step 1: Compute $\mathbf{u}_1$ and normalize

$$\mathbf{u}_1 = \mathbf{v}_1 = (1,1,0), \quad \|\mathbf{u}_1\| = \sqrt{2}, \quad \mathbf{e}_1 = \frac{1}{\sqrt{2}}(1,1,0).$$

3.2.2 Step 2: Orthogonalize $\mathbf{v}_2$ against $\mathbf{u}_1$

$$\text{proj}_{\mathbf{u}_1}(\mathbf{v}_2) = \frac{\mathbf{v}_2 \cdot \mathbf{u}_1}{\mathbf{u}_1 \cdot \mathbf{u}_1}\mathbf{u}_1 = \frac{1}{2}(1,1,0).$$

$$\mathbf{u}_2 = \mathbf{v}_2 - \text{proj}_{\mathbf{u}_1}(\mathbf{v}_2) = (1,0,1) - \left(\tfrac{1}{2}, \tfrac{1}{2}, 0\right) = \left(\tfrac{1}{2}, -\tfrac{1}{2}, 1\right).$$

$$\|\mathbf{u}_2\| = \sqrt{\tfrac{1}{4} + \tfrac{1}{4} + 1} = \sqrt{\tfrac{3}{2}} = \frac{\sqrt{6}}{2}, \quad \mathbf{e}_2 = \frac{\mathbf{u}_2}{\|\mathbf{u}_2\|} = \frac{1}{\sqrt{6}}(1,-1,2).$$

3.2.3 Step 3: Orthogonalize $\mathbf{v}_3$ against $\mathbf{u}_1, \mathbf{u}_2$

$$v_3 = (0,1,1), \quad u_1 = (1,1,0), \quad u_2 = \left(\tfrac{1}{2}, -\tfrac{1}{2}, 1\right).$$

$$\mathrm{proj}_{u_1}(v_3) = \frac{v_3 \cdot u_1}{u_1 \cdot u_1} u_1 = \frac{1}{2}(1,1,0) = \left(\tfrac{1}{2}, \tfrac{1}{2}, 0\right).$$

$$\mathrm{proj}_{u_2}(v_3) = \frac{v_3 \cdot u_2}{u_2 \cdot u_2} u_2 = \frac{\frac{1}{2}}{\frac{3}{2}} u_2 = \frac{1}{3}\left(\tfrac{1}{2}, -\tfrac{1}{2}, 1\right) = \left(\tfrac{1}{6}, -\tfrac{1}{6}, \tfrac{1}{3}\right).$$

$$u_3 = v_3 - \mathrm{proj}_{u_1}(v_3) - \mathrm{proj}_{u_2}(v_3) = \left(-\tfrac{2}{3}, \tfrac{2}{3}, \tfrac{2}{3}\right).$$

$$\|u_3\| = \frac{2}{\sqrt{3}}, \qquad e_3 = \frac{u_3}{\|u_3\|} = \left(-\tfrac{1}{\sqrt{3}}, \tfrac{1}{\sqrt{3}}, \tfrac{1}{\sqrt{3}}\right).$$

3.3 Resulting Orthonormal Basis

$$\mathbf{e}_1 = \tfrac{1}{\sqrt{2}}(1,1,0), \quad \mathbf{e}_2 = \tfrac{1}{\sqrt{6}}(1,-1,2), \quad \mathbf{e}_3 = (-\tfrac{5}{6}, \tfrac{5}{6}, \tfrac{1}{3}).$$

3.4 Interpretation

Through the Gram–Schmidt process, we have transformed the original linearly independent set $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ into an orthonormal basis $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ for the same subspace.

3.5 Why Orthogonal Bases Simplify Computations

1. **Simple Coordinate Computation:** As illustrated in example 1, finding the coordinates of a vector $\mathbf{v}$ requires solving a system of n equations. In an *orthonormal basis*, each coordinate is obtained directly by a dot product:

   $$a_i = \mathbf{v} \cdot \mathbf{u}_i.$$

   This reduces the computational cost from $O(n^3)$ to $O(n^2)$.

2. **Simplified Matrix Inversion:** If A is an orthogonal matrix (columns form an orthonormal basis), then

   $$A^{-1} = A^T.$$

   Thus, solving $A\mathbf{x} = \mathbf{b}$ becomes $\mathbf{x} = A^T\mathbf{b}$, which requires only matrix–vector multiplication instead of full inversion.